

Research Assistant API

API Reference Documentation

v0.1.0

FastAPI service for scholarly discovery, metadata analytics,
semantic similarity, and user research workflows
over the Leeds OpenAlex subset.

Base URL: `https://research-assistant-api.up.railway.app`

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Overview

Introduction

The Research Assistant API provides programmatic access to a curated subset of the Leeds OpenAlex scholarly corpus. It exposes endpoints for paper search and discovery, author and topic lookup, citation graph traversal, semantic similarity, corpus analytics, and user-owned research workflow management (projects, reading lists, and annotations).

Base URL

```
https://research-assistant-api.up.railway.app
```

Authentication

Protected endpoints use **OAuth2 Password Bearer** tokens. Obtain a token from `POST /auth/register` or `POST /auth/login` and include it in every authenticated request:

```
Authorization: Bearer <access_token>
```

Endpoints that require authentication are marked **Auth Required**.

Content Type

All request and response bodies use `application/json`.

Pagination

List endpoints support pagination via `limit` and `offset` query parameters.

Parameter	Type	Required	Description
<code>limit</code>	integer	No	Number of results to return. Range: 1–100. Default varies by endpoint.
<code>offset</code>	integer	No	Number of results to skip. Default: 0.

Error Responses

Validation errors return **HTTP 422** (`HTTPValidationError`). See the Schemas section for the full definition. Example structure:

```
{
  "detail": [
    {
      "loc": ["body", "email"],
      "msg": "field required",
      "type": "value_error.missing",
      "input": null,
      "ctx": {}
    }
  ]
}
```

Health

Service health and liveness endpoints.

Health Check

GET /health Return a lightweight liveness response for the API service.

Responses

200 Returns object `(string → string)` — e.g. `{"status":"ok"}`

Authentication

User registration, login, and bearer-token identity.

Register User — POST /auth/register

POST /auth/register Create a user account and return a bearer access token.

Request Body — UserRegistrationRequest (required)

Parameter	Type	Required	Description
email	string	Yes	User email address. Length: 3–255 characters.
password	string	Yes	Account password. Length: 8–255 characters.

Responses

201 Account created. Returns AccessTokenResponse.

422 Validation error. Returns HTTPValidationError.

Log In User — POST /auth/login

POST /auth/login Authenticate a user and return a bearer access token.

Request Body — UserLoginRequest (required)

Parameter	Type	Required	Description
email	string	Yes	Registered email address. Length: 3–255 characters.
password	string	Yes	Account password. Length: 8–255 characters.

Responses

200 Login successful. Returns AccessTokenResponse.

422 Validation error. Returns HTTPValidationError.

Get Current User — GET /auth/me

GET /auth/me **Auth Required** Return the authenticated user linked to the bearer token.

Responses

200 Returns AuthenticatedUser.

422 Validation error. Returns HTTPValidationError.

Analytics

Corpus-level analytics, top-paper ranking, and publication trends.

List Top Papers — GET /analytics/top-papers

GET /analytics/top-papers Rank papers by citation count with optional topic and year filters.

Query Parameters

Parameter	Type	Required	Description
topic	string	No	Filter by topic name.
year	integer	No	Filter by publication year (≥ 0).
limit	integer	No	Results per page. Range: 1–100. Default: 20.
offset	integer	No	Pagination offset. Default: 0.

Responses

200 Returns array of AnalyticsPaperItem, ordered by descending citation count.

422 Validation error. Returns HTTPValidationError.

Get Topic Distribution — GET /analytics/topics

GET /analytics/topics Summarise topic coverage and aggregate citation counts across the corpus.

Query Parameters

Parameter	Type	Required	Description
year	integer	No	Filter by publication year (≥ 0).
limit	integer	No	Results per page. Range: 1–100. Default: 50.
offset	integer	No	Pagination offset. Default: 0.

Responses

200 Returns array of TopicAnalyticsItem.

422 Validation error. Returns HTTPValidationError.

Get Publication Trends — GET /analytics/trends

GET /analytics/trends Summarise publication output and citation totals by year.

Query Parameters

Parameter	Type	Required	Description
start_year	integer	No	Earliest year to include (≥ 0).
end_year	integer	No	Latest year to include (≥ 0).

Responses

200 Returns array of PublicationTrendItem, ordered by year.

422 Validation error. Returns HTTPValidationError.

Papers

Paper discovery, detail, similarity, citations, and paper-scoped annotations.

Search Papers — GET /papers/search

GET /papers/search Search the local Leeds corpus by free text, topic, year, and citation threshold.

Query Parameters

Parameter	Type	Required	Description
query	string	No	Free-text search query.
topic	string	No	Filter by topic name.
year	integer	No	Filter by publication year (≥ 0).
citation_count	integer	No	Minimum citation count threshold (≥ 0).
limit	integer	No	Results per page. Range: 1–100. Default: 20.
offset	integer	No	Pagination offset. Default: 0.

Responses

200 Returns array of PaperSummary.

422 Validation error. Returns HTTPValidationError.

Get Paper — GET /papers/{paper_id}

GET /papers/{paper_id} Return a paper record with metadata, authorship, topic, and citation count.

Path Parameters

Parameter	Type	Required	Description
paper_id	string	Yes	Unique paper identifier.

Responses

200 Returns PaperDetail.

422 Validation error. Returns HTTPValidationError.

List Similar Papers — GET /papers/{paper_id}/similar

GET /papers/{paper_id}/similar Return semantically similar papers using stored embeddings.

Path Parameters

Parameter	Type	Required	Description
paper_id	string	Yes	Source paper identifier.

Query Parameters

Parameter	Type	Required	Description
limit	integer	No	Results per page. Range: 1–100. Default: 10.
offset	integer	No	Pagination offset. Default: 0.

Responses

200 Returns array of SimilarPaperResponse, ordered by descending similarity score.

422 Validation error. Returns HTTPValidationError.

Get Citation Neighbourhood — GET /papers/{paper_id}/citations**GET** /papers/{paper_id}/citations Return papers cited by, and citing, the requested paper within the Leeds citation subgraph.

Path Parameters

Parameter	Type	Required	Description
paper_id	string	Yes	Central paper identifier.

Query Parameters

Parameter	Type	Required	Description
limit	integer	No	Results per page. Range: 1–100. Default: 20.
offset	integer	No	Pagination offset. Default: 0.

Responses

200 Returns CitationNeighborhoodResponse.**422** Validation error. Returns HTTPValidationError.**Get Citation Path** — GET /papers/{paper_id}/path/{target_paper_id}**GET** /papers/{paper_id}/path/{target_paper_id} Find a directed shortest citation path within the Leeds citation subgraph.

Path Parameters

Parameter	Type	Required	Description
paper_id	string	Yes	Source paper identifier.
target_paper_id	string	Yes	Target paper identifier.

Query Parameters

Parameter	Type	Required	Description
max_depth	integer	No	Maximum traversal depth. Range: 1–12. Default: 6.

Responses

200 Returns CitationPathResponse.**422** Validation error. Returns HTTPValidationError.**Create Paper Annotation** — POST /papers/{paper_id}/annotations**POST** /papers/{paper_id}/annotations **Auth Required** Create a private annotation for the authenticated user on a paper.

Path Parameters

Parameter	Type	Required	Description
paper_id	string	Yes	Paper to annotate.

Request Body — AnnotationCreateRequest (required)

Parameter	Type	Required	Description
text	string	Yes	Annotation content. Length: 1–5000 characters.

Responses

201 Annotation created. Returns AnnotationResponse.**422** Validation error. Returns HTTPValidationError.

List Paper Annotations — GET /papers/{paper_id}/annotations**GET** /papers/{paper_id}/annotations **Auth Required** List the authenticated user's annotations for a paper.

Path Parameters

Parameter	Type	Required	Description
<code>paper_id</code>	string	Yes	Paper identifier.

Query Parameters

Parameter	Type	Required	Description
<code>limit</code>	integer	No	Results per page. Range: 1–100. Default: 20.
<code>offset</code>	integer	No	Pagination offset. Default: 0.

Responses

200 Returns array of AnnotationResponse.**422** Validation error. Returns HTTPValidationError.

Annotations

Private annotation read, update, and delete operations.

Get Annotation — GET /annotations/{annotation_id}

GET /annotations/{annotation_id} **Auth Required** Fetch a single private annotation owned by the authenticated user.

Path Parameters

Parameter	Type	Required	Description
annotation_id	string	Yes	Annotation identifier.

Responses

200 Returns AnnotationResponse.

422 Validation error. Returns HTTPValidationError.

Update Annotation — PATCH /annotations/{annotation_id}

PATCH /annotations/{annotation_id} **Auth Required** Update the text of a private annotation owned by the authenticated user.

Path Parameters

Parameter	Type	Required	Description
annotation_id	string	Yes	Annotation identifier.

Request Body — AnnotationUpdateRequest (required)

Parameter	Type	Required	Description
text	string	Yes	Updated annotation content. Length: 1–5000 characters.

Responses

200 Returns updated AnnotationResponse.

422 Validation error. Returns HTTPValidationError.

Delete Annotation — DELETE /annotations/{annotation_id}

DELETE /annotations/{annotation_id} **Auth Required** Delete a private annotation owned by the authenticated user.

Path Parameters

Parameter	Type	Required	Description
annotation_id	string	Yes	Annotation identifier.

Responses

204 Annotation deleted. No response body.

422 Validation error. Returns HTTPValidationError.

Authors

Author search, detail, and paper lookup endpoints.

List Authors — GET /authors

GET /authors List authors in the local Leeds corpus subset.

Query Parameters

Parameter	Type	Required	Description
limit	integer	No	Results per page. Range: 1–100. Default: 50.
offset	integer	No	Pagination offset. Default: 0.

Responses

200 Returns array of AuthorDetail.

422 Validation error. Returns HTTPValidationError.

Search Authors — GET /authors/search

GET /authors/search Search authors in the local Leeds corpus subset by name or author ID.

Query Parameters

Parameter	Type	Required	Description
query	string	Yes	Search term — name or author ID. Minimum length: 1.
limit	integer	No	Results per page. Range: 1–100. Default: 50.
offset	integer	No	Pagination offset. Default: 0.

Responses

200 Returns array of AuthorDetail.

422 Validation error. Returns HTTPValidationError.

Get Author — GET /authors/{author_id}

GET /authors/{author_id} Return author metadata and affiliation details.

Path Parameters

Parameter	Type	Required	Description
author_id	string	Yes	Author identifier.

Responses

200 Returns AuthorDetail.

422 Validation error. Returns HTTPValidationError.

List Author Papers — GET /authors/{author_id}/papers

GET /authors/{author_id}/papers List papers associated with a specific author.

Path Parameters

Parameter	Type	Required	Description
author_id	string	Yes	Author identifier.

Query Parameters

Parameter	Type	Required	Description
<code>limit</code>	integer	No	Results per page. Range: 1–100. Default: 20.
<code>offset</code>	integer	No	Pagination offset. Default: 0.

Responses

- 200** Returns array of PaperSummary.
- 422** Validation error. Returns HTTPValidationError.

Topics

Topic lookup and topic-scoped paper listing.

List Topics — GET /topics

GET /topics List topics available in the local Leeds corpus subset.

Query Parameters

Parameter	Type	Required	Description
limit	integer	No	Results per page. Range: 1–100. Default: 50.
offset	integer	No	Pagination offset. Default: 0.

Responses

200 Returns array of TopicListItem.

422 Validation error. Returns HTTPValidationError.

List Topic Papers — GET /topics/{topic_id}/papers

GET /topics/{topic_id}/papers List papers assigned to a topic.

Path Parameters

Parameter	Type	Required	Description
topic_id	string	Yes	Topic identifier.

Query Parameters

Parameter	Type	Required	Description
limit	integer	No	Results per page. Range: 1–100. Default: 20.
offset	integer	No	Pagination offset. Default: 0.

Responses

200 Returns array of PaperSummary.

422 Validation error. Returns HTTPValidationError.

Projects

User-owned projects, reading lists, and recommendation workflows.

List Projects — GET /projects

GET /projects **Auth Required** List projects owned by the authenticated user.

Responses

200 Returns array of ProjectResponse.

Create Project — POST /projects

POST /projects **Auth Required** Create a user-owned research project.

Request Body — ProjectCreateRequest (required)

Parameter	Type	Required	Description
title	string	Yes	Project title. Length: 1–255 characters.
description	string	No	Optional project description. Maximum 5000 characters.

Responses

201 Project created. Returns ProjectResponse.

422 Validation error. Returns HTTPValidationError.

Get Project — GET /projects/{project_id}

GET /projects/{project_id} **Auth Required** Return a single project owned by the authenticated user.

Path Parameters

Parameter	Type	Required	Description
project_id	string	Yes	Project identifier.

Responses

200 Returns ProjectResponse.

422 Validation error. Returns HTTPValidationError.

Update Project — PATCH /projects/{project_id}

PATCH /projects/{project_id} **Auth Required** Update the title or description of a user-owned project.

Path Parameters

Parameter	Type	Required	Description
project_id	string	Yes	Project identifier.

Request Body — ProjectUpdateRequest (required)

Parameter	Type	Required	Description
title	string	No	Updated project title. Length: 1–255 characters.
description	string	No	Updated project description. Maximum 5000 characters.

Responses

200 Returns updated ProjectResponse.

422 Validation error. Returns HTTPValidationError.

Delete Project — DELETE /projects/{project_id}

DELETE /projects/{project_id} **Auth Required** Delete a user-owned project and its dependent workflow records.

Path Parameters

Parameter	Type	Required	Description
project_id	string	Yes	Project identifier.

Responses

- 204** Project deleted. No response body.
- 422** Validation error. Returns HTTPValidationError.

Add Reading-List Item — POST /projects/{project_id}/reading-list

POST /projects/{project_id}/reading-list **Auth Required** Add a paper to a project reading list.

Path Parameters

Parameter	Type	Required	Description
project_id	string	Yes	Project identifier.

Request Body — ReadingListItemCreateRequest (required)

Parameter	Type	Required	Description
paper_id	string	Yes	Paper to add. Length: 1–255 characters.
priority	string	No	Reading priority: low, medium, or high. Default: medium.
notes	string	No	Optional notes. Maximum 5000 characters.

Responses

- 201** Item added. Returns ReadingListItemResponse.
- 422** Validation error. Returns HTTPValidationError.

List Reading-List Items — GET /projects/{project_id}/reading-list

GET /projects/{project_id}/reading-list **Auth Required** List reading-list items for a user-owned project.

Path Parameters

Parameter	Type	Required	Description
project_id	string	Yes	Project identifier.

Responses

- 200** Returns array of ReadingListItemResponse.
- 422** Validation error. Returns HTTPValidationError.

List Project Recommendations — GET /projects/{project_id}/recommendations

GET /projects/{project_id}/recommendations **Auth Required** Recommend papers from reading-list context using semantic, citation, or hybrid scoring.

Path Parameters

Parameter	Type	Required	Description
project_id	string	Yes	Project identifier.

Query Parameters

Parameter	Type	Required	Description
<code>mode</code>	string	No	Scoring mode: <code>semantic</code> , <code>citation</code> , or <code>hybrid</code> . Default: <code>hybrid</code> .
<code>semantic_weight</code>	number	No	Weight for semantic scoring. Range: 0.0–1.0.
<code>citation_weight</code>	number	No	Weight for citation scoring. Range: 0.0–1.0.
<code>limit</code>	integer	No	Results per page. Range: 1–100. Default: 10.
<code>offset</code>	integer	No	Pagination offset. Default: 0.

Responses

200 Returns array of `ProjectRecommendationResponse`.

422 Validation error. Returns `HTTPValidationError`.

Reading List

Reading-list item update and delete operations.

Update Reading-List Item — PATCH /reading-list-items/{item_id}

PATCH /reading-list-items/{item_id} **Auth Required** Update reading-list priority or notes for an item owned by the authenticated user.

Path Parameters

Parameter	Type	Required	Description
<code>item_id</code>	string	Yes	Reading-list item identifier.

Request Body — ReadingListItemUpdateRequest (required)

Parameter	Type	Required	Description
<code>priority</code>	string	No	Updated priority: <code>low</code> , <code>medium</code> , or <code>high</code> .
<code>notes</code>	string	No	Updated notes. Maximum 5000 characters.

Responses

200 Returns updated ReadingListItemResponse.

422 Validation error. Returns HTTPValidationError.

Delete Reading-List Item — DELETE /reading-list-items/{item_id}

DELETE /reading-list-items/{item_id} **Auth Required** Delete a reading-list item owned by the authenticated user.

Path Parameters

Parameter	Type	Required	Description
<code>item_id</code>	string	Yes	Reading-list item identifier.

Responses

204 Item deleted. No response body.

422 Validation error. Returns HTTPValidationError.

Schemas

Complete data model definitions for all request bodies, response types, and error structures. All schemas map directly to the OpenAPI component definitions.

UserRegistrationRequest

Field	Type	Req.	Description
<code>email</code>	string	Yes	User email address. Length: 3–255 characters.
<code>password</code>	string	Yes	Account password. Length: 8–255 characters.

UserLoginRequest

Field	Type	Req.	Description
<code>email</code>	string	Yes	Registered email address. Length: 3–255 characters.
<code>password</code>	string	Yes	Account password. Length: 8–255 characters.

AccessTokenResponse

Field	Type	Req.	Description
<code>access_token</code>	string	Yes	Bearer token for authenticated requests.
<code>token_type</code>	string	No	Token scheme. Always "bearer".
<code>user</code>	AuthenticatedUser	Yes	The authenticated user record.

AuthenticatedUser

Field	Type	Req.	Description
<code>id</code>	string	Yes	User UUID.
<code>email</code>	string	Yes	User email address.
<code>created_at</code>	string (date-time)	Yes	Account creation timestamp (ISO 8601).

PaperSummary

Base paper record returned by list and search endpoints.

Field	Type	Req.	Description
<code>id</code>	string	Yes	Paper identifier.
<code>title</code>	string	Yes	Paper title.
<code>publication_year</code>	integer	Yes	Year of publication.
<code>publication_date</code>	string (date)	No	Full publication date (ISO 8601).
<code>citation_count</code>	integer	Yes	Total citation count.
<code>doi</code>	string	No	Digital Object Identifier.
<code>journal</code>	string	No	Journal or venue name.
<code>language</code>	string	No	Language code (e.g. "en").
<code>work_type</code>	string	No	Work type (e.g. "article", "book-chapter").
<code>topic</code>	TopicSummary	No	Assigned topic.

PaperDetail

Full paper record returned by GET `/papers/{paper_id}`. Includes all *PaperSummary* fields plus the following.

Field	Type	Req.	Description
<code>id</code>	string	Yes	Paper identifier. <i>(from PaperSummary)</i>
<code>title</code>	string	Yes	Paper title. <i>(from PaperSummary)</i>
<code>publication_year</code>	integer	Yes	Year of publication. <i>(from PaperSummary)</i>
<code>publication_date</code>	string (date)	No	Full publication date. <i>(from PaperSummary)</i>
<code>citation_count</code>	integer	Yes	Total citation count. <i>(from PaperSummary)</i>
<code>doi</code>	string	No	Digital Object Identifier. <i>(from PaperSummary)</i>
<code>journal</code>	string	No	Journal or venue name. <i>(from PaperSummary)</i>
<code>language</code>	string	No	Language code. <i>(from PaperSummary)</i>
<code>work_type</code>	string	No	Work type. <i>(from PaperSummary)</i>
<code>topic</code>	TopicSummary	No	Assigned topic. <i>(from PaperSummary)</i>
<code>abstract</code>	string	No	Paper abstract.
<code>authors</code>	PaperAuthorSummary[]	Yes	Ordered list of authors.

PaperAuthorSummary

Field	Type	Req.	Description
<code>id</code>	string	Yes	Author identifier.
<code>name</code>	string	Yes	Author display name.
<code>orcid</code>	string	No	ORCID identifier.
<code>institution</code>	InstitutionSummary	No	Affiliated institution.
<code>author_position</code>	integer	No	Position in author list (1-indexed).
<code>is_corresponding</code>	boolean	Yes	Whether this is the corresponding author.

SimilarPaperResponse

Returned by GET `/papers/{paper_id}/similar`. Includes all *PaperSummary* fields plus:

Field	Type	Req.	Description
<code>id</code>	string	Yes	Paper identifier. <i>(from PaperSummary)</i>
<code>title</code>	string	Yes	Paper title. <i>(from PaperSummary)</i>
<code>publication_year</code>	integer	Yes	Year of publication. <i>(from PaperSummary)</i>
<code>publication_date</code>	string (date)	No	Full publication date. <i>(from PaperSummary)</i>
<code>citation_count</code>	integer	Yes	Total citation count. <i>(from PaperSummary)</i>
<code>doi</code>	string	No	Digital Object Identifier. <i>(from PaperSummary)</i>
<code>journal</code>	string	No	Journal or venue name. <i>(from PaperSummary)</i>
<code>language</code>	string	No	Language code. <i>(from PaperSummary)</i>
<code>work_type</code>	string	No	Work type. <i>(from PaperSummary)</i>
<code>topic</code>	TopicSummary	No	Assigned topic. <i>(from PaperSummary)</i>
<code>similarity_score</code>	number	Yes	Cosine similarity score in range [0, 1].

CitationNeighborhoodResponse

Field	Type	Req.	Description
<code>paper_id</code>	string	Yes	Central paper identifier.
<code>cited_papers</code>	PaperSummary[]	Yes	Papers cited by this paper.
<code>citing_papers</code>	PaperSummary[]	Yes	Papers that cite this paper.

CitationPathResponse

Field	Type	Req.	Description
<code>source_paper_id</code>	string	Yes	Source paper identifier.
<code>target_paper_id</code>	string	Yes	Target paper identifier.
<code>path_length</code>	integer	Yes	Number of hops in the shortest path.
<code>path</code>	PaperSummary[]	Yes	Ordered list of papers forming the citation path (source to target).

AuthorDetail

Field	Type	Req.	Description
<code>id</code>	string	Yes	Author identifier.
<code>name</code>	string	Yes	Author display name.
<code>orcid</code>	string	No	ORCID identifier.
<code>institution</code>	InstitutionSummary	No	Affiliated institution.
<code>paper_count</code>	integer	Yes	Number of papers in the corpus for this author.

InstitutionSummary

Field	Type	Req.	Description
<code>id</code>	string	Yes	Institution identifier.
<code>name</code>	string	Yes	Institution name.
<code>country</code>	string	No	ISO country code.

TopicSummary

Field	Type	Req.	Description
<code>id</code>	string	Yes	Topic identifier.
<code>name</code>	string	Yes	Topic name.
<code>field</code>	string	No	Broad research field.

TopicListItem

Includes all *TopicSummary* fields plus:

Field	Type	Req.	Description
<code>id</code>	string	Yes	Topic identifier. <i>(from TopicSummary)</i>
<code>name</code>	string	Yes	Topic name. <i>(from TopicSummary)</i>
<code>field</code>	string	No	Broad research field. <i>(from TopicSummary)</i>
<code>paper_count</code>	integer	Yes	Number of papers assigned to this topic.

AnalyticsPaperItem

Field	Type	Req.	Description
<code>id</code>	string	Yes	Paper identifier.
<code>title</code>	string	Yes	Paper title.
<code>publication_year</code>	integer	Yes	Year of publication.
<code>citation_count</code>	integer	Yes	Total citation count.
<code>journal</code>	string	No	Journal or venue name.
<code>topic</code>	TopicSummary	No	Assigned topic.

TopicAnalyticsItem

Field	Type	Req.	Description
<code>id</code>	string	Yes	Topic identifier.
<code>name</code>	string	Yes	Topic name.
<code>field</code>	string	No	Broad research field.
<code>paper_count</code>	integer	Yes	Number of papers in this topic.
<code>total_citation_count</code>	integer	Yes	Sum of citations across all papers in this topic.
<code>average_citation_count</code>	number	Yes	Mean citation count per paper in this topic.

PublicationTrendItem

Field	Type	Req.	Description
<code>publication_year</code>	integer	Yes	Calendar year.
<code>paper_count</code>	integer	Yes	Number of papers published that year.
<code>total_citation_count</code>	integer	Yes	Sum of all citations for papers from that year.
<code>average_citation_count</code>	number	Yes	Mean citation count per paper for that year.

AnnotationCreateRequest

Field	Type	Req.	Description
<code>text</code>	string	Yes	Annotation content. Length: 1–5000 characters.

AnnotationUpdateRequest

Field	Type	Req.	Description
<code>text</code>	string	Yes	Updated annotation content. Length: 1–5000 characters.

AnnotationResponse

Field	Type	Req.	Description
<code>id</code>	string	Yes	Annotation identifier.
<code>user_id</code>	string	Yes	Owning user identifier.
<code>paper_id</code>	string	Yes	Annotated paper identifier.
<code>text</code>	string	Yes	Annotation content.

Field	Type	Req.	Description
<code>created_at</code>	string (date-time)	Yes	Creation timestamp (ISO 8601).

ProjectCreateRequest

Field	Type	Req.	Description
<code>title</code>	string	Yes	Project title. Length: 1–255 characters.
<code>description</code>	string	No	Optional project description. Maximum 5000 characters.

ProjectUpdateRequest

Field	Type	Req.	Description
<code>title</code>	string	No	Updated project title. Length: 1–255 characters.
<code>description</code>	string	No	Updated project description. Maximum 5000 characters.

ProjectResponse

Field	Type	Req.	Description
<code>id</code>	string	Yes	Project identifier.
<code>user_id</code>	string	Yes	Owning user identifier.
<code>title</code>	string	Yes	Project title.
<code>description</code>	string	No	Project description.
<code>created_at</code>	string (date-time)	Yes	Creation timestamp (ISO 8601).

ProjectRecommendationResponse

Returned by `GET /projects/{project_id}/recommendations`. Includes all *PaperSummary* fields plus scoring metadata.

Field	Type	Req.	Description
<code>id</code>	string	Yes	Paper identifier. <i>(from PaperSummary)</i>
<code>title</code>	string	Yes	Paper title. <i>(from PaperSummary)</i>
<code>publication_year</code>	integer	Yes	Year of publication. <i>(from PaperSummary)</i>
<code>publication_date</code>	string (date)	No	Full publication date. <i>(from PaperSummary)</i>
<code>citation_count</code>	integer	Yes	Total citation count. <i>(from PaperSummary)</i>
<code>doi</code>	string	No	Digital Object Identifier. <i>(from PaperSummary)</i>
<code>journal</code>	string	No	Journal or venue name. <i>(from PaperSummary)</i>
<code>language</code>	string	No	Language code. <i>(from PaperSummary)</i>
<code>work_type</code>	string	No	Work type. <i>(from PaperSummary)</i>
<code>topic</code>	TopicSummary	No	Assigned topic. <i>(from PaperSummary)</i>
<code>recommendation_score</code>	number	Yes	Final composite recommendation score.
<code>semantic_score</code>	number	Yes	Semantic similarity component score.
<code>citation_score</code>	number	Yes	Citation graph component score.
<code>scoring_mode</code>	string (enum)	Yes	Mode used: <code>semantic</code> , <code>citation</code> , or <code>hybrid</code> .
<code>semantic_weight</code>	number	Yes	Weight applied to the semantic score.
<code>citation_weight</code>	number	Yes	Weight applied to the citation score.

ReadingListItemCreateRequest

Field	Type	Req.	Description
<code>paper_id</code>	string	Yes	Paper to add. Length: 1–255 characters.
<code>priority</code>	string (enum)	No	Reading priority: <code>low</code> , <code>medium</code> , or <code>high</code> . Default: <code>medium</code> .
<code>notes</code>	string	No	Optional notes. Maximum 5000 characters.

ReadingListItemUpdateRequest

Field	Type	Req.	Description
<code>priority</code>	string (enum)	No	Updated priority: <code>low</code> , <code>medium</code> , or <code>high</code> .
<code>notes</code>	string	No	Updated notes. Maximum 5000 characters.

ReadingListItemResponse

Field	Type	Req.	Description
<code>id</code>	string	Yes	Reading-list item identifier.
<code>project_id</code>	string	Yes	Parent project identifier.
<code>paper_id</code>	string	Yes	Paper identifier.
<code>priority</code>	string (enum)	Yes	Priority level: <code>low</code> , <code>medium</code> , or <code>high</code> .
<code>notes</code>	string	No	User notes for this item.
<code>created_at</code>	string (date-time)	Yes	Creation timestamp (ISO 8601).
<code>paper</code>	PaperSummary	Yes	Embedded full paper summary.

HTTPValidationError

Returned on HTTP 422 Unprocessable Entity responses.

Field	Type	Req.	Description
<code>detail</code>	ValidationError[]	No	Array of individual validation failure records.

ValidationError

A single field-level validation failure within `HTTPValidationError.detail`.

Field	Type	Req.	Description
<code>loc</code>	(string integer)[]	Yes	Location path of the failed field — e.g. <code>["body", "email"]</code> .
<code>msg</code>	string	Yes	Human-readable error message.
<code>type</code>	string	Yes	Machine-readable error type identifier.
<code>input</code>	any	No	The value that failed validation (may be null).
<code>ctx</code>	object	No	Additional context about the validation failure.